The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia

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We estimate the aggregate productivity gains from reducing barriers to internal labor migration in Indonesia, accounting for worker selection and spatial differences in human capital. We distinguish between movement costs, which mean workers will move only if they expect higher wages, and amenity differences, which mean some locations must pay more to attract workers. We find modest but important aggregate impacts. We estimate a 22 percent increase in labor productivity from removing all barriers. Reducing migration costs to the US level, a high-mobility benchmark, leads to a 7.1 percent productivity boost. These figures hide substantial heterogeneity. The origin population that benefits most sees a 104 percent increase in average earnings from a complete barrier removal, or a 25 percent gain from moving to the US benchmark.

I. Introduction

Recent evidence suggests that a policy of encouraging internal labor migration could have large productivity effects in developing countries. On the macro side, Gollin, Lagakos, and Waugh (2014) show that nonagricul-

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tural (urban) workers produce four times more than their agricultural (rural) counterparts. On the micro side, Bryan, Chowdhury, and Mobarak (2014) show a 33 percent increase in consumption from experimentally induced seasonal migration. Neither of these results, however, is definitive: The experimental estimates apply only to seasonal migration, and to a specific part of Bangladesh. The macro estimates do not account for selection on unobservables (Young 2013), and only apply to movement between rural and urban areas.

This paper uses micro data from Indonesia to quantify the aggregate effect of increasing mobility. Two observations motivate our approach. First, migration could increase productivity if it (1) allows individuals to sort into a location in which they are personally more productive (sorting), (2) allows more people to live in more productive locations (agglomeration), or (3) both.[1]

Second, in the absence of constraints or amenity differentials, people will maximize their production; therefore, a policy that encourages migration will have no effect on output if there are no existing constraints on mobility.

We build a model in which workers have idiosyncratic location-specific productivity and in which locations differ in their overall productivity. This setup allows for both sorting and agglomeration effects. Into this framework we incorporate two kinds of mobility constraints. Movement costs exist if workers must be paid higher wages to induce them to work away from home. Compensating wage differentials exist if workers must be paid higher wages to work in low-amenity locations. The result is a general equilibrium Roy model in which workers sort across locations that have heterogeneous amenities and productivities. The model is similar to that used by Hsieh et al. (2019); our approach also has close connections to the seminal work of Hsieh and Klenow (2009).[2] We use this structural framework

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[1] We use the term agglomeration to encompass two mechanisms that are often separated in the literature: the first is more people living in locations with higher fundamental productivity; the second is the externalities that arise when more people live close to each other.

[2] Our framework also has much in common with recent quantitative models of economic geography such as Allen and Arkolakis (2014), Redding (2016), and Desmet, Nagy, and Rossi-Hansberg (2018). We also draw on important contributions studying commuting, e.g., Monte et al. (2018) and Ahlfeldt et al. (2015). Our framework is similar to that used in work by Tombe and Zhu (2019). Relative to that paper, we use more detailed micro data that enable us to directly estimate the extent of selection, and we are interested in a different set of questions.

to quantify the change in aggregate productivity that would result from removing movement costs and/or equalizing amenity differentials. Like Hsieh and Klenow (2009) and Caselli (2005), we do not consider specific policies but rather try to quantify the potential impacts of a set of policy options.

Our main contribution is combining this quantitative framework with rich micro data from Indonesia. The Indonesian data, which are unique in recording location of birth, current location, and current earnings, allow for particularly transparent identification of key model parameters. For example, we are able to identify the key parameter that controls sorting from a simple linear regression of the origin-destination wage on the origin-destination migration share. Intuitively, across-destination/within-origin variation in migration rates can be used to estimate the strength of selection forces, but few data sets contain the information necessary to run this regression.

Before turning to our structural analysis, we document five motivational facts, which suggest both that movement costs and compensating differentials exist and that selection is important in the data. Our rich micro data allow us to demonstrate these facts. In the case of movement costs, we first show that a gravity relationship holds in the data. A 10 percent reduction in the distance between two locations leads to a 7 percent increase in the proportion of migrants who flow between the two locations. We also show that people who live farther from their location of birth have higher wages. A doubling of distance leads to a 3 percent increase in average wages, suggesting that people need to be compensated to induce them to move away from home. In running these regressions, we think of distance as a proxy for movement costs, which may not capture all policy-relevant constraints. For compensating differentials, we show that workers in observably low-amenity locations receive higher wages.

Selection effects also appear to be important in the data: the greater the share of people born in origin o that move to destination d, the lower their average wage. The elasticity of average wage with respect to share is approximately -0.04. Importantly, because our model is one in which movement costs reduce migration and lead to selection, we show that there is almost no effect of distance on average wages once the proportion of the origin population at the destination is controlled for; proportion migrating is sufficient to account for the wage differences. All these effects are predicted by our model. We also show that the same set of motivating facts holds for migration between states in the United States.

To estimate the potential effects of policy, we turn to our structural model. When estimating the model, we treat both movement costs and amenity differentials as nonparametric objects to be inferred from the data. Movement costs are nonparametric in the sense that we estimate a separate cost for each origin-destination pair that is independent of dis-

tance or any other measure. Our measures of movement costs therefore capture a wide range of barriers. For example, language differences that reduce bilateral migration would be a movement cost. Amenities, following the tradition in urban economics, are estimated as a residual.

The choice to treat movement costs and amenity differentials in this way reflects our view that amenities are hard to measure and distance is unlikely to capture all policy-relevant dimensions of movement costs.

Our model allows for straightforward quantification of the effects of reducing movement-cost-driven or amenity-driven wage differentials. The intuition is straightforward. We first generate counterfactual population distributions by estimating where people would live if we removed their empirical tendency to stay at their place of birth and their tendency to avoid some locations that have high measured productivity. Next, we ask how productivity would change if people moved as suggested by our counterfactuals. Our model of selection implies that each additional migrant will earn less than the last; to account for this we need to understand how wages change as workers move. Since selection, in our model, is relative to location of birth, it is the average wages of people from a given origin who live in a given destination that matter. As noted above, our unique data, which capture both location of birth and current location of work, combined with an instrumental-variables strategy inspired by our model, allow us to estimate the relevant elasticity.

Our results suggest moderate aggregate gains but important heterogeneity. Removing all frictions is predicted to increase aggregate productivity by 22 percent. These gains are modest relative to the potential gains suggested by studies such as Gollin et al. (2014), but are in line with what one may expect from other microeconomic studies. For the people born in some locations, however, the results are much larger, with predicted gains peaking at 104 percent. We show, theoretically and empirically, that gains are larger for origins that have higher dispersion in average wages across destinations. Because complete barrier removal may be impossible, we also compute the gains from moving to the US level of movement costs, which we see as a high-mobility benchmark. We predict an aggregate productivity boost of 7.1 percent, with the origin that gains most seeing a 25 percent increase. We conclude that while migration that improves the static allocation of labor is unlikely to have very large productivity effects of the sort estimated, for example, by Hsieh and Klenow (2009), targeted policies may have big impacts on the lives of some communities.

Our paper differs from existing approaches in three ways. First, we consider region-to-region rather than rural-to-urban movement. Since Lewis (1954) and Harris and Todaro (1970), the development-and-migration literature has been dominated by rural-to-urban studies. In our setting this is potentially inappropriate. Figure 1 shows kernel density plots of the log of the average monthly wage, calculated at the subprovince (Indonesian re-

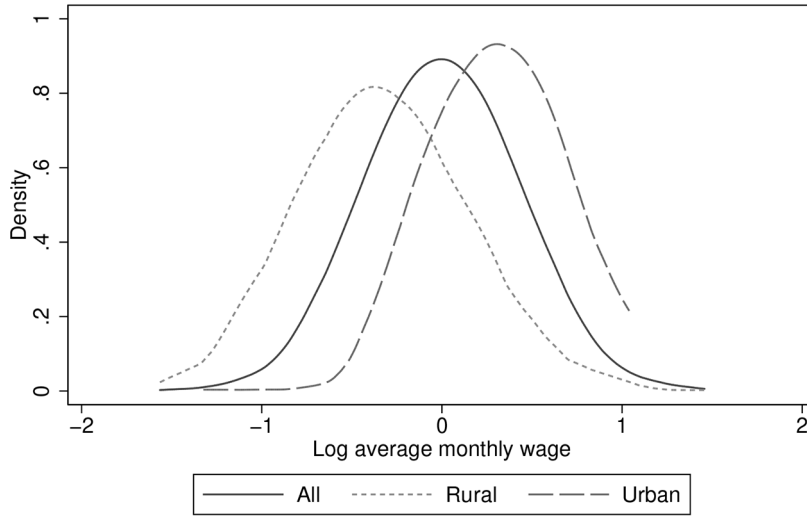


FIG. 1.—Distribution of wages: regency level. Log average monthly wage is demeaned of year fixed effects. Unit of observation is the regency. Regency is defined as either rural or urban to match the national share of rural. Sources: 1995 SUPAS, 2011 SUSENAS, 2012 SUSENAS.

gency) level and broken down by rural/urban status.[3] The figure highlights that while there is large variation between regencies, there is little overall variation between rural and urban locations. Table 1 shows that the majority of migration also occurs within category, rather than across category: between 75 and 85 percent of migration out of urban areas is to another urban area, and between 25 and 30 percent of migration out of a rural area is to another rural area. Focusing only on rural-urban migration misses the within-rural and the within-urban migrations.

Second, we focus on counterfactual estimates that predict the effect of removing constraints. While we can learn much from work documenting returns to past migration,[4] there are challenges moving from these estimates to predictions of future returns. On one hand, selection effects mean future migrants may earn less than past migrants; on the other hand, migration policies work by reducing constraints, and so will tend to encourage migration where past movement was minimal. Because of this, past returns

[3] We code regencies that have greater than median rural population share as rural, and the remaining regencies as urban. Appendix fig. 1 shows that the same patterns hold if we plot the distribution of individual, rather than regency average, wages.

[4] Recent work by Kleemans and Magruder (2018), Hicks et al. (2017), Beegle, de Weerdt, and Dercon (2011), and Garlick, Leibbrandt, and Levinsohn (2016) provide important estimates of the returns to, and impact of, past migrations in Indonesia, Kenya, Tanzania, and South Africa, respectively.

TABLE 1
MIGRATION RATES BY ORIGIN FOR INDONESIA

	Rural	Urban	All
1995:			
Migration rate	32.3	35.8	33.7
Moves within category	31.1	74.6	49.4
2011:			
Migration rate	38.7	33.7	35.7
Moves within category	24.4	84.2	58.7
2012:			
Migration rate	38.9	34.1	35.8
Moves within category	25.4	83.8	60.7

SOURCES.—1995 SUPAS, 2011 SUSENAS, 2012 SUSENAS.

NOTE.—Migration is measured as living in a regency other than the birth regency. Regencies are classified as rural or urban on the basis of the share of their population that reports being rural; we choose the cutoff to classify the regency as rural to match the national urbanization rate for each year.

may contain little information on the likely effects of future policies. For our analysis we directly estimate the impact of removing constraints. Our only use of past migration is to estimate the strength of selection effects. While this approach is similar to macroeconomic estimates based on productivity gaps (e.g., Gollin et al. 2014), it accounts for selection effects that are likely to be important.

Finally, we take account of general equilibrium effects. First, by incorporating sorting, we allow for aggregate productivity gains in the absence of large net population flows. Second, we calibrate agglomeration, congestion, rental, and price elasticities using consensus estimates, and we then assess how our results depend on these parameters.

Our results are limited in three ways. First, we look only at static gains, leaving examination of dynamic effects for future work.[5] Second, when doing our counterfactuals we look only at the productivity impacts and only at gains. We do not consider welfare effects of removing migration restrictions (which may be negative), and we do not consider the costs of policy. A full consideration of costs is difficult and can be avoided if benefits are small. Third, we do not consider specific policies but rather provide estimates of the total gains that may be available. Our approach is similar, therefore, to the development accounting and macro misallocation literatures (Caselli 2005; Hsieh and Klenow 2009).

The paper starts by laying out five motivational facts. These facts strongly suggest that spatial labor markets in Indonesia are characterized by costs

[5] There are several potential sources of dynamic gains. For example, migration costs may be endogenous (Carrington, Detragiache, and Vishwanath 1996), firm openings may depend on the pool of available migrant labor, or both.

of movement, compensating wage differentials, and selection on productivity. The facts imply the possibility of productivity gains from increased movement. We then provide a simple two-location example that explains how we quantify the possible gains. We follow this by briefly describing our formal model, discussing identification and estimation, and demonstrating that our structurally estimated parameters correlate sensibly with real-world proxy measures. Finally, we present results from counterfactual exercises.

II. Data, Motivation, and Two-Location Example

A. Data

Our approach has specific data requirements. In our view, people will only migrate if their earnings increase enough to compensate them for living away from home (which we take to be their location of birth). We therefore need data that record an individual's location of birth, current location of work, and earnings. Our interest in aggregate returns implies that data have to be geographically representative. Because we want to nonparametrically estimate movement costs, the data set must be large enough that it records flows between all pairs of locations. Data of this kind are available in very few locations, and Indonesia is the unique country that meets these specifications and has location recorded at a level below the equivalent of a state.

Our Indonesian data come from the 1995 SUPAS (Intercensal Population Survey) and from the 2011 and 2012 SUSENAS (National Socioeconomic Survey). These data sets record, for a large representative set of people, location of birth (origin o), current work location d (which could be the same as the origin), and monthly earnings (which we refer to as the wage). A limitation of these data is that they do not capture earnings for the self-employed. To understand the biases that this may introduce, we supplement the SUPAS/SUSENAS data with data from the Indonesia Family Life Survey (IFLS), a longitudinal survey. The IFLS has a much smaller sample and by design covers only 13 out of 25 Indonesian provinces, but it does collect more detailed information on incomes, including for the self-employed, and follows the same individuals over time. While we cannot use the IFLS data to estimate the structural model, we can use it to understand how key parameter estimates are affected by the limitations of the SUPAS/SUSENAS data. We also use data from the United States, both to show that our migration facts hold more generally and to generate a suitable counterfactual for a high-mobility economy. We use the 1990 five-percent census sample and the 2010 American Community Survey, as these dates overlap most closely with our Indonesia dates. In all

cases, we restrict the sample to be male heads of household between 15 and 65 years old.[6] Summary statistics for the Indonesian and US samples are given in appendix table 1; summary statistics for the IFLS sample are given in appendix table 2 (the appendixes are available as an online supplement). All wage variables are reported in monthly terms.

In the United States, we have locations of birth and work recorded at the level of the state; in Indonesia, we have this information for the regency (and, aggregating up, at the province level).[7] Because of the census nature of our data, our measure of migration is permanent migration based on a repeated cross section. This may miss people who have moved multiple times or who have moved and returned home. To ascertain the scope of these issues, we look at detailed migration histories collected in the IFLS. A migration episode in the IFLS is defined as a move lasting at least six months. We find that multiple and return migrations are not large issues in our context. As appendix table 3 shows, migration in Indonesia can be broadly characterized as one permanent migration episode, made in adulthood. Looking at male household heads, conditional on moving out of the birth province, 69 percent of all migrants make only one migration, 26 percent make two moves, and only 5 percent make three or more moves. Importantly, only 8 percent of migration is undertaken by people under the age of 16, and 50 percent of second migrations are made by people returning home. These numbers are broadly similar to those for the United States, where Kennan and Walker (2011) find that the average male migrant makes 1.98 moves and 50.2 percent of movers move home.

We use the 2005 and 2011 Village Potential Statistics (PODES) data sets to get measures of amenity. These data are reported by a local leader and contain information on all locations, both urban and rural, in Indonesia. We collapse to the regency level, using population weights.

B. Five Empirical Facts about Migration

From our data, we can calculate the proportion of people from each origin o that move to each destination d, which we denote π_{do}, as well as the

[6] This restriction reduces our sample size in Indonesia from 419,760 to 187,065. We restrict to male heads of households as our model is one in which migration is motivated by work, and women and children may migrate for a more diverse set of reasons. As we discuss below, the key parameter that drives our estimates of the gains from migration is the distribution of talent in the population. Reassuringly, estimates of this key parameter change little when we include both non–household heads and women. Tables available upon request.

[7] A regency is a second-level administrated subdivision below a province and above a district. For all surveys, we drop the provinces of Papua and West Papua. We generate a set of regencies that have maintained constant geographical boundaries between 1995 and 2010. This primarily involves merging together regencies that were divided in 2001. This leaves us with a sample of 281 regencies. Later, for the structural estimates we aggregate regencies up to the level of province, of which there are 25.

average wage within origin-destination pair, $\overline{\text{wage}}_{do}$. Using these data, we document five empirical facts about migration in Indonesia. We present these five facts at the regency level. For the later estimation of the model, we aggregate regencies into provinces.[8] We then show that these basic facts about migration also hold true in the US sample.

FACT 1 (Gravity: Movement costs affect location choice). Controlling for origin and destination fixed effects, the share of people born in o who move to d is decreasing in the distance between o and d.

To document fact 1, we run a regression

$$\ln(\pi_{dot}) = \delta_{dt} + \delta_{ot} + \beta \ln(\text{dist}_{do}) + \epsilon_{dot},$$

where δ_{dt} and δ_{ot} are destination-year and origin-year fixed effects, respectively, and dist_{do} is the straight distance between regency o and regency d.[9] The destination effect controls for any productivity or amenity differences across destinations, and the origin effect controls for the benefits of other possible locations from the perspective of those living at the origin (this term is similar to the multilateral-resistance term in the trade literature).

We interpret distance as a proxy for movement costs, which we think include both the costs of travel as well as a broader set of concerns including cultural differences and language differences. The results are shown in table 2, column 1. We estimate that the elasticity of π_{do} with respect to dist_{do} is negative, strongly significant, and sizeable. A 10 percent increase in distance leads to a 7 percent reduction in the proportion migrating. These results suggest that there are costs of moving people across space.

FACT 2 (Movement costs create productivity wedges). Controlling for origin and destination fixed effects, the average wage of people born in origin o and living in destination d is increasing in the distance between o and d.

To establish fact 2, we run the regression

$$\ln(\overline{\text{wage}}_{dot}) = \delta_{dt} + \delta_{ot} + \beta \ln(\text{dist}_{do}) + \epsilon_{dot}.$$

The results are shown in table 2, column 2. We estimate that the elasticity of the average wage with respect to distance is positive, strongly significant, and sizeable. A doubling of the distance between origin and desti-

[8] The Indonesian results are also robust to aggregating to the province level (appendix table 4) and using the IFLS data (appendix table 5). We report our motivational facts at the regency level because this increases power. When we conduct our structural estimation we aggregate to the province level to reduce the number of zeros in the bilateral migration matrix. We discuss the IFLS results in more detail in Sec. VI.E.2, where we consider the robustness of our estimates.

[9] The term dist_{do} is the straight-line distance, in kilometers, between the centroid of regency o and the centroid of regency d. We have experimented with movement time, generated using Dijkstra's algorithm and assumptions about the time cost of different types of travel. This does not materially affect the results.

TABLE 2
FIVE FACTS ABOUT MIGRATION IN INDONESIA

	Movement Costs		Selection		Compensating Differential
Dependent Variable	log π_{odt} (1)	log w_{odt} (2)	log w_{odt} (3)	log w_{odt} (4)	log w_{odt} (5)
Log distance	−.717	.029		.007	
	(.009)***	(.001)***		(.002)***	
Log share migrating			−.039	−.031	
			(.001)***	(.003)***	
Amenities					−.023
					(.010)**
Destination × year fixed effects	Yes	Yes	Yes	Yes	No
Origin × year fixed effects	Yes	Yes	Yes	Yes	Yes
Destination fixed effects					Yes
Number of individuals	187,065	186,763	186,763	186,763	185,357
Number of region pairs	25,540	25,244	25,244	25,244	25,050

Sources.—1995 SUSENAS, 2011 SUSENAS, 2012 SUSENAS.

Note.—Here log w_{odt} is the log of the share of population migrating from o to d in year t; log w_{odt} is the log of the average wage of migrants from origin o in destination d in time t. An observation is an origin-destination regency pair. Amenity measure is the standardized value of the first principal component. Two-way clustering of standard errors at the origin-year and destination-year reported in cols. 1–4. Clustered standard errors, at the level of the origin-year, reported in col. 5. Number of observations changes between columns because not all pairs with positive migration flows have observed wages.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

nation leads to a 3 percent increase in the average wages. These impacts can be very large. For example, the straight-line distance from Denpasar to Jakarta on the western tip of Java is about 1000 km. On the other hand, the distance from Denpasar to Banyuwangi on the eastern tip of Java is about 100 km. Our estimates suggest that the average wage of migrants from Denpasar to Jakarta will be 30 percent more than those to Banyuwangi.

As we explain in more detail in our two-location example below, this fact suggests that movement costs reduce productivity. To easily illustrate this, consider two locations d and d' that are identical except that d is closer to o. Fact 2 implies that those who choose to move to d' have higher average wages than those who choose to move to d. Under the hypotheses that the two destinations are identical, that workers are rational, and that workers are paid their marginal product, the only way that those in d' can have higher wages is if distance (movement costs) dissuaded the moves of some positive-productivity movers, who would have earned less than the current average wage.

Fact 3 (Selection). Controlling for origin and destination fixed effects, the elasticity of average wages with respect to origin population share is negative.

Fact 3 is documented by running the regression

$$\ln(\overline{\text{wage}}_{dot}) = \delta_{ot} + \delta_{dt} + \beta \ln(\pi_{dot}) + \epsilon_{dot}. \quad (1)$$

Estimates from this regression are presented in table 2, column 3. Our estimates, which are strongly statistically significant, show that the elasticity of average wages is negative. In Indonesia, a doubling of the share of people who migrate to a particular destination leads to a 4 percent decrease in average wages. This fact suggests selection on productivity. If workers are paid their marginal products, then, controlling for destination productivity, the only way that average wages can differ across destinations within origin is if the distribution of worker skills is a function of π_{do}. We show below that the coefficient on $\ln \pi_{do}$ in this regression is the key parameter that measures the importance of selection and sorting in our model. This fact is subject to a potential endogeneity concern: any shock to productivity in destination d that differentially affects people from different origins o will tend to also alter π_{do}. Below, we use our full theoretical model to motivate an instrument to correct for this. Instrumentation changes the quantitative results but does not alter the qualitative fact.

Fact 4 (Movement costs reduce productivity by reducing selection). The elasticity of average wage to distance drops to almost zero after controlling for the fraction of the origin population that migrate.

We document fact 4 by running the regression

$$\ln(\overline{\text{wage}}_{dot}) = \delta_{ot} + \delta_{dt} + \beta \ln(\pi_{dot}) + \gamma \ln(\text{dist}_{do}) + \epsilon_{dot}. \quad (2)$$

Results are presented in table 2, column 4. The coefficient on $\ln \pi_{dt}$ changes little when the distance control is added, but the magnitude of the estimated distance effect, while still positive and statistically signficant, drops relative to the results in column 2, falling to an economically insignificant size.

Facts 3 and 4 together suggest a framework in which increasing movement costs, proxied here by distance, lead to a reduction in the proportion of people who move (fact 1). This, in turn, leads to an increase in wages (fact 2), but these wage effects are generated by a selection effect created by a reduced proportion moving (facts 3 and 4). This is consistent with our discussion of facts 2 and 3, where we assume that workers are paid their marginal productivity, so once destination and origin fixed effects are controlled for, wage differences reflect selection. Importantly, fact 4 suggests that our structural approach of estimating the impact of reducing movement costs using the elasticity of wage with respect to proportion moving will capture most of the effects of removing movement cost.

Fact 5 (Compensating wage differentials). Controlling for origin fixed effects, locations with higher amenities have lower wages.

To document fact 5 we run the regression

$$\ln(\overline{\text{wage}}_{dot}) = \delta_{ot} + \delta_{dt} + \beta\,\text{amen}_{dt} + \epsilon_{dot},$$

where amen_{dt} is measured amenity in destination d at time t. To determine amenity, we take six different measures of amenity from the Indonesian PODES survey and convert to a single measure by taking the first principal component.[10] We then standardize to give this variable a zero mean and unit standard deviation. The results are shown in table 2, column 5. Our estimates imply that a 1 standard deviation increase in amenities leads to a 2.3 percent decrease in average wages. This is direct evidence that firms pay a compensating wage differential to attract workers to low-amenity locations. Importantly, there is little endogeneity concern with the sign of this result. While one may be concerned that higher-wage locations can afford higher amenities, this result goes in the opposite direction.

The basic facts also hold in the US data.—Table 3 shows that the main facts also hold for the United States, when migration is defined as crossing a state border. We show evidence for the first four facts as we do not have a measure of amenity at the state level for the United States. Starting with column 1, we find evidence of a gravity equation for migration. Column 2 shows that wages in the destination are increasing in the distance measured. Column 3 shows that wages in the destination are decreasing in the share of people migrating, and column 4 shows that the wage effect is driven by the share of people migrating, not the distance effect. This implies that the same framework can be used to interpret migration patterns in the United States: increasing movement costs, proxied here by distance, lead to a reduction in the proportion of people that move, which, because of selection effects, leads to an increase in wages.

C. *An Example with Two Locations*

In this section we briefly discuss a two-location version of our model. We highlight the mechanisms through which migration costs and amenity differentials reduce productivity. We also show how we estimate the productivity impacts of policies that reduce migration frictions. Because of

[10] We have two broad categories of amenities: amenities affecting services ("ease" amenities), such as the ease of reaching a hospital, and negative amenities affecting pollution ("pollution" amenities), such as the presence of water pollution in the last year. A full list of the amenities in the data is given in appendix table 6. For the motivating fact we use the "ease" amenities only because we are concerned that pollution is picking up economic output directly. We use the first principle component because we are interested in computing a unidimensional measure of amenities. We only require our measure to be a proxy measure for amenities.

TABLE 3
FOUR FACTS ABOUT MIGRATION IN THE UNITED STATES

	Movement Costs		Selection	
Dependent Variable	log π_{odt} (1)	log w_{odt} (2)	log w_{odt} (3)	log w_{odt} (4)
Log distance	−.553	.023		−.004
	(.018)***	(.002)***		(.004)
Log share migrating			−.043	−.050
			(.003)***	(.006)***
Destination × year fixed effects	Yes	Yes	Yes	Yes
Origin × year fixed effects	Yes	Yes	Yes	Yes
Destination fixed effects				
Number of individuals	2,294,054	2,294,046	2,294,046	2,294,046
Number of region pairs	5,084	5,076	5,076	5,076

Sources.—1990 census, 2010 American Community Survey.

Note.—Here log π_{odt} is the log of the share of population migrating from o to d in year t; log w_{odt} is the log of the average wage of migrants from origin o in destination d in time t. An observation is an origin-destination state pair. Two-way clustering of standard errors at the origin-year and destination-year reported in cols. 1–4. Number of observations changes between columns because not all pairs with positive migration flows have observed wages.

*** Significant at the 1 percent level.

the simplicity of the two-location model, we can give an intuitive graphical analysis.

We think of each work place, or destination d, as being characterized by a productivity w_d and amenity α_d. We also assume that each location produces different goods and that people's productivities depend on their location. In particular, we assume that the wage of person i living in destination d is $w_d s_{id}$, where s_{id} is the skill level of person i for location d. Total utility for person i from location o who decides to live and work in destination d is then $\alpha_d w_d s_{id}(1 - \tau_{do})$, where τ_{do} is the cost that a person born in origin o pays to live in destination d. We refer to τ_{do} either as a movement cost or as a migration cost. We assume that $\tau_{do} \in [0, 1]$, $\tau_{oo} = 0$, and $\tau_{do} = \tau_{od}$. In our empirical work we will back out α_d and τ_{do} as residuals, and so this way of writing the utility function normalizes the measure of amenities and movement costs relative to wages.

Figure 2 shows the distribution of skill (s_{id}) across two locations, which we call A and B; the figure is drawn from the perspective of people born in location B. If there were no frictions, people would live where their earnings, $w_d s_{id}$, are highest. As drawn, location A has the higher productivity, and all those above the ray OE, which has slope w_B/w_A, should move to location A (i.e., those in regions I, II, and III should migrate). If the two locations had equal productivity, those above the 45 degree line (in areas I and II) should have moved to maximize productivity.

With movement costs, people from B must be compensated for their move to A. This means that earnings in A are effectively less valuable, and only those above the line OC, which has slope $w_B/w_A(1 - \tau_{AB})$, will

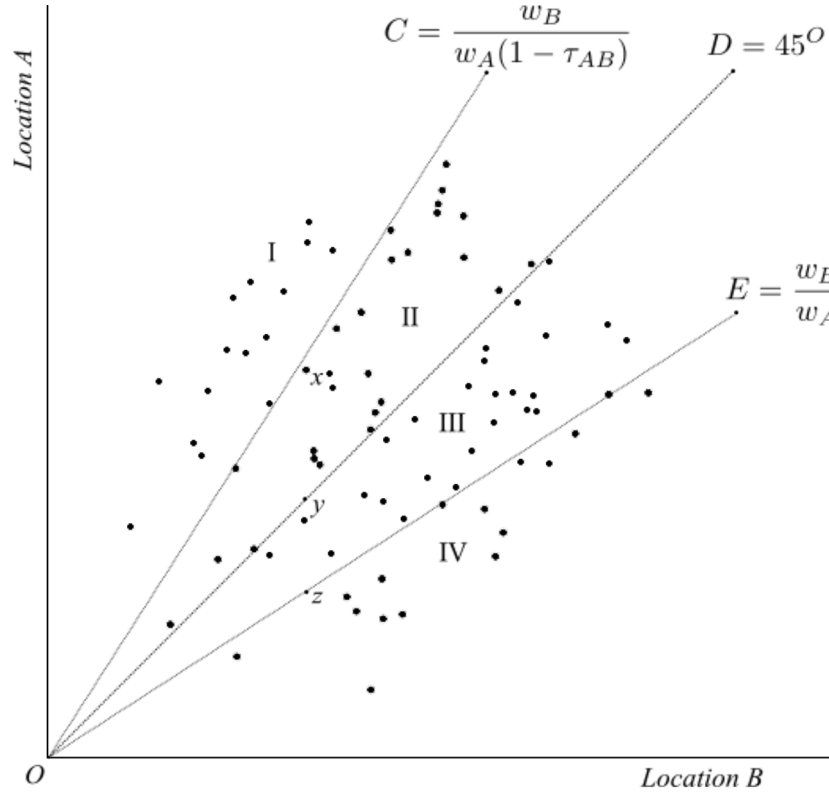


FIG. 2.—Productivity and location choices of people born in location *B*

choose to move. We can divide those born in location *B* into four groups. Those below ray *OE* (the dots in region *IV*) should not move, because their returns are highest to stay in *B*, and they do not. Those above *OE* and below the 45 degree line (the dots in region *III*) should move, because *A* has higher productivity than *B*. The higher productivity in *A* compensated these people for the fact that their comparative advantage lies in *B*. With movement costs, these people do not move. Those above the 45 degree line and below ray *OC* (the dots in region *II*) should move, for two reasons. First, they have a comparative advantage in location *A*. Second, *A* is a more productive location. Consider person *x*: she loses productivity equal to the distance *xy* because she has a comparative advantage in *A* but does not move, and an additional amount *yz* because *A* is more productive. These two channels mean that movement costs reduce productivity by reducing sorting, and by reducing agglomeration in high-productivity locations. Finally, those above *OC* in region *I* should move and they always do. In line with all models inspired by the work of Roy

(1951), this figure shows that those with the most to gain will move first, and therefore suggests limits on the gains to promoting migration. It also highlights that most of the gains from migration are to be had by encouraging movement to places where costs are high, and so historical movements have been low.

Fact 2 and its interpretation can be seen in this diagram. As movement costs increase, fewer people move to A and the wages of those that move increase. This increase occurs because some people who would have been more productive in A now choose to stay in B.[11]

Amenities also move worker locations away from the productivity-maximizing allocation. With amenities but no movement costs, people now maximize $\alpha_d w_d s_{id}$. The effect can be understood in the same diagram. With no movement costs and B having higher relative amenity, the ray OC would have slope $\alpha_B w_B / \alpha_A w_A$. The same effects—a lack of sorting and too little agglomeration—are present, and, so long as the level of amenity in A differs from the level of amenity in B, productivity will not be maximized. The main difference between amenity differentials and movement costs is that movement costs will reduce migration relative to home, while amenity differentials reduce the number of people living in one location relative to the other.

It is worth noting that selection plays two roles in our model. On one hand, worker heterogeneity and selection are a source of gains. Movement costs, which stop workers from moving to their location of comparative advantage, reduce productivity. On the other hand, selection limits the potential gains from moving more workers to high-productivity locations. In the absence of selection on productivity, all workers who move will have the same wage, and so aggregate impacts of removing amenity differentials can be larger.

Our empirical task is to estimate the gain in productivity that would come from allocating people to their productivity-maximizing location. This problem can be separated into two parts. First, we estimate the movement response. This is equivalent to estimating how many people lie in the triangle OCE. This is conceptually straightforward. In the case in which there are no productivity differences between locations, the productivity-maximizing choice is that half the people from B will stay in B and half will live in A. Second, we estimate how this movement will affect the average wages of the four groups in our data: those from A that move to B, those from B that live in A, and those that stay in A or B. Functional form assumptions laid out below imply that average wages for these groups are

[11] This fact depends on the properties of the skill distribution. In the language of Lagakos and Waugh (2013), comparative and absolute advantages must be aligned. App. D discusses the relationship between comparative and absolute advantages in our framework. We find evidence consistent with comparative and absolute advantages being aligned. See also Adao (2016) for a discussion.

a constant elasticity function of the fraction of the origin population that live in the destination. This elasticity is estimable given our data, which records origin and destination, and is shown in fact 3 above. Because our data record the proportion of people from each origin who live in each destination π_{do}, and counterfactual population distributions can be expressed in the same way, this elasticity is sufficient to estimate the counterfactual aggregate productivity. In the next two sections, we lay out how these ideas extend to more than two locations, how to account for heterogeneous location productivities, and how we incorporate general equilibrium effects.

III. Model

In this section we present a static general equilibrium model of migration. The model is designed to be as simple as possible; we discuss a number of extensions and how they might affect the results in appendix B. The model is an adaptation of the labor-sorting model in Hsieh et al. (2019), which itself draws on Eaton and Kortum (2002). The model also has similarities with recent work on quantitative economic geography, particularly Allen and Arkolakis (2014), and quantitative urban economics, particularly Monte, Redding, and Rossi-Hansberg (2018) and Ahlfeldt et al. (2015).[12]

The economy consists of N locations. Workers are born in a particular origin (o), draw a skill for each destination (d), and sort across destinations according to wages, amenities, and migration costs. Migration costs are relative to the birth location. Wages and amenities are endogenous and adjust to ensure equilibrium. We first discuss how workers choose where to live and work taking wages and amenities as given, and then turn to production and general equilibrium determination of wages and amenities.

A. *Utility and Sorting*

Here L_o individuals are born in each origin o. Each person i receives a skill draw s_{id} for each possible work destination $d \in N$. It seems unlikely that this is literally true; what we have in mind is that people have different talents for different industries, and that different destinations have different represented industries. So, for example, a person who is very talented at data science would have a high draw for San Francisco, while someone with a talent for banking would have a relatively high draw for New York.[13]

[12] The urban models include a cost of commuting, which is conceptually similar to our treatment of movement costs. See Redding and Rossi-Hansberg (2017) for a review of work on quantitative spatial models.

[13] In fact, as noted by Lagakos and Waugh (2013), the assumption that talent is drawn from a Fréchet distribution is consistent with this interpretation. Hence, we can think of the assumption that s_d is drawn from a Fréchet distribution as being consistent with a richer

The individual also receives a skill draw for her location of origin. Skill is drawn from a multivariate Fréchet distribution,

$$F(s_1,\ldots,s_N) = \exp\left(-\left\{\sum_{d=1}^{N} s_d^{-[\tilde{\theta}/(1-\rho)]}\right\}^{1-\rho}\right),$$

which does not depend on the location of birth.[14] Here $\tilde{\theta}$ measures the extent of skill dispersion or the importance of comparative advantage. As $\tilde{\theta}$ decreases, there is a greater difference between skills across locations. The term ρ measures the correlation in skills across locations. As ρ increases, individuals with a high draw in destination d are also likely to have a high draw for destination d'. The interpretation is that each different location has a different set of required skills. To the extent that $\tilde{\theta}$ is estimated to be high, locations do not differ greatly in their skill requirements. We allow for correlation between skill draws to allow for general talent, and the case in which talent is unidimensional is a limiting case as $\rho \to 1$. Throughout it is useful to work with $\theta = \tilde{\theta}/(1-\rho)$ rather than $\tilde{\theta}$.

Innate skills are combined with schooling in the location of origin to become human capital. Location d human capital for individual i born in location o is given by

$$h_{ido} = s_{id} q_o.$$

Throughout, we refer to q_o as the quality of schooling in o, but it likely reflects a broader set of factors that contribute to human capital. We consider the possibility of endogenous acquisition of human capital in appendix B. The wage per effective unit of labor in destination d for someone from origin o is given by $w_d \epsilon^w_{do}$, where w_d is destination d productivity, and ϵ^w_{do} is a mean 1 lognormally distributed error that captures any reason why people from origin o may be more productive in destination d (i.e., it is an origin-specific labor demand shifter in destination d). We assume that the error is observed by the individual before migrating, and we introduce it because it allows for a meaningful discussion (in Sec. IV.A) of an intuitively important endogeneity issue: any unmeasured characteristic

setting in which individuals receive skill draws for a large number of industries in each destination, and choose the industry that maximizes their wage. The main challenge to this interpretation is that data limitations mean that we are forced to assume that talent draws for each destination are drawn from the same Fréchet distribution; we show in app. D that there is no evidence that the shape parameters differ by destination or origin, consistent with this assumption. Given this interpretation of the shock, migration frictions will include frictions that prevent people from moving industry, if that industry move requires migration.

[14] We later introduce a difference in skill by origin, q_o; the resulting model is isomorphic to one in which the scale parameter of the Fréchet parameter differs across locations. The important assumption is that $\tilde{\theta}$ does not differ by origin.

that increases productivity in destination d will also increase movement to destination d. The wage for individual i from origin o is therefore

$$\text{wage}_{ido} = w_d \epsilon_{do}^{w} h_{ido} = w_d \epsilon_{do} s_{id} q_o.$$

Indirect utility for individual i from origin o living in destination d is given by

$$U_{ido} = \alpha_d \epsilon_{do}^{\alpha} (1 - \tau_{do}) w_d \epsilon_{do}^{w} s_{id} q_o \equiv \bar{w}_{do} s_{id}. \tag{3}$$

The term $w_d \epsilon_{do} q_o s_{id}$ captures consumption, which is equal to the wage. The term α_d measures the amenity of location d and captures the need for compensating differentials. Moving to a location with half the amenity level would be compensated for by a doubling of earnings. Amenities could include natural beauty, the availability of services, or rental rates.[15] The term ϵ_{do}^{α} is assumed to be mean zero and lognormally distributed; it captures differences in amenity that depend on location of origin. Again, this error term is observed by the individual before making the decision to move, and ensures that the model does not perfectly fit the data. The term τ_{do} captures the utility cost of living away from home (the origin o), and we refer to it as a moving cost. We assume that $\tau_{oo} = 0$, so that moving away from home to a destination d would require an individual to be compensated with $(1 - \tau_{do})$ times the income. For example, compared to consumption at the origin o, the same level of consumption at destination d may be less pleasurable as it is not undertaken with family and friends. We assume throughout that movement costs are symmetric, so that $\tau_{do} = \tau_{od}$. With this background, known results regarding the Fréchet distribution imply the following results.

First, let π_{do} be the portion of people from origin o who choose to work in destination d. We have

$$\pi_{do} = \frac{\tilde{w}_{do}^{\theta}}{\sum_{j=1}^{N} \tilde{w}_{jo}^{\theta}}, \tag{4}$$

where $\tilde{w}_{do} = w_d \epsilon_{do}^{w} \alpha_d \epsilon_{do}^{\alpha} (1 - \tau_{do})$. Here $\tilde{w}_{do}$ measures the attractiveness of location d for someone from o. Equation (4) is the key sorting equation, and it asserts that sorting depends on relative returns, relative amenities, and relative movement costs; it does not depend on the quality of human capital formation in the origin, q_o. That sorting does not depend on q_o is key to our exercise: we wish to distinguish between human capital or schooling effects that lead to higher production and human capital effects that are a barrier to migration. Barriers to migration coming from

[15] Much work in the tradition of Rosen (1979) and Roback (1982) separates out rents from other amenities. We discuss how to incorporate rents in app. B.3.

differences in human capital are, to the extent that they are symmetric, captured in τ_{do}. To the extent that human capital differences are a barrier to migration but are not symmetric, they will be captured in ϵ^{w}_{do} and will not form part of our counterfactuals.

Second, we can use this characterization to determine the average skill of workers from o working in d by noting that

$$E(s_d|\text{choose } d) = \pi_{do}^{-(1/\theta)}\bar{\Gamma}, \tag{5}$$

where $\bar{\Gamma} = \Gamma(1 - [1/\theta(1 - \rho)])$ and $\Gamma(\cdot)$ is the Gamma function. This equation implies that the more people from o move to d, the lower is their average skill. This is intuitive as it implies that there is less selection: the marginal migrant is drawn from further down the left tail of the talent distribution. Finally, we can work out the average wage in a particular location for people from a given origin:

$$\overline{\text{wage}}_{do} = w_d \epsilon_{do} q_o E(s_d|\text{choose } d) = w_d \epsilon^{w}_{do} q_o \pi_{do}^{-(1/\theta)}\bar{\Gamma}. \tag{6}$$

Equations (4) and (6) are our main estimating equations. Taking logs of these two equations also shows that the model is consistent with the motivating facts discussed earlier. Fact 1, gravity, is an estimate of equation (4), where distance is substituted for moving cost. Facts 2 and 5 come from (6), with π_{do} substituted from equation (4). Facts 3 and 4 come directly from (6).

One important implication of our modeling choices is worth noting. When we observe large average wage gaps between locations or sectors, it is tempting to think that there will be large productivity gains to moving people. Our model highlights two reasons why this may not be the case. First, the gaps may reflect selection, as in Young (2013). Second, those in low-productivity locations may simply have low human capital in total, captured by low q_o in our model. In our empirical work, we will estimate q_o, allowing for unobservable heterogeneity in the quality of human capital production.

B. *Production and General Equilibrium*

Each location is assumed to produce a differentiated good y_d. This output is produced by a large number of firms in each location that each produce an identical product according to a linear production technology. Profits for firm j in location d are given by

$$\Pi_{jd} = p_d A_d h_{jd} - w_{jd} h_{jd},$$

where A_d is labor productivity in location d, p_d is the price, which firms take as given, w_{jd} is the wage paid by firm j, and h_{jd} is the total amount of human capital employed by firm j. Firms compete for laborers by set-

ting wages w_{jd}, which implies that in equilibrium $w_{jd} = w_d$ and $\Pi_{jd} = 0 \ \forall j$, and so

$$w_d = p_d A_d.$$

Total economy-wide production is given by the constant elasticity of substitution aggregate

$$Y = \left(\sum_{d=1}^{N} y_d^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)},$$

where y_d is the total production in location d, and σ captures the degree of substitutability between products produced by different locations.[16] Prices p_d are determined by assuming that a representative firm chooses y_d to maximize total economy output less the costs of production $\Sigma_d p_d y_d$.[17] This aggregate final good is costlessly traded across the country, and is chosen as the numeraire. Utility is linear in the consumption of the aggregate final good, leading to the utility function given in (3).

We allow productivity to be endogenous. Total output of good d depends on the amount of human capital in location d according to the function

$$y_d = A_d H_d,$$

where H_d is the total human capital (or effective labor units) available at location d and

$$A_d = \bar{A}_d H_d^{\gamma}$$

is the productivity of location d. In this formulation, $\bar{A}_d$ can be thought of as intrinsic productivity—an exogenous parameter—which may change over time. For example, New York may presently have high productivity due to its proximity to a port, but this may have been even more important 100 years ago. Current labor productivity, A_d, depends on intrinsic productivity and the total amount of human capital in location d, with γ parameterizing the extent of human capital spillovers, or productive agglomeration externalities.

Finally, amenity is also endogenously determined. We assume

$$\alpha_d = \bar{\alpha}_d \hat{L}_d^{\lambda},$$

where $\bar{\alpha}_d$ is baseline amenity; for example, natural beauty, λ, is a measure of congestion effects and likely to be less than zero, and $\hat{L}_d$ is the (endogenously determined) population of location d.

[16] If $\sigma \to \infty$ all products are perfect substitutes, so the case in which all locations produce the same good is a limit case of our model. An alternative specification would be to allow for locations to produce goods that are perfectly substitutable with a decreasing returns to scale production function. Hsieh and Moretti (2019) show that the two approaches are isomorphic.

[17] This implies that prices are determined by the equation $p_d = (Y/q_d)^{1/\sigma}$.

It is important to note one key characteristic of the model. Dividing through (4) and (6), it is easy to show

$$\frac{\overline{\text{wage}}_{do}}{\overline{\text{wage}}_{d'o}} = \left(\frac{\alpha_{d'}}{\alpha_d}\right)\left(\frac{1-\tau_{d'o}}{1-\tau_{do}}\right).$$

Hence, within origin, there are no wage gaps (per unit of human capital) without frictions (or, if only migration frictions are removed, then there are no amenity-adjusted wage gaps).[18] There are two key assumptions that drive this result. The first assumption is that comparative and absolute advantages are aligned. This leads to the fact that reducing frictions will lead to a convergence in wages. The second assumption is that the elasticity of wages to the proportion of the population (from an origin) is constant and is the same across all locations. In our model we assume a Fréchet distributional assumption that hard-bakes assumption 1, and then because we assume that shape parameters are constant across all locations, this leads to assumption 2. We discuss these points fully in appendix D, where we argue that it is not possible to reject these two assumptions in the data.

The fact that, within origin, there are no wage gaps without frictions means that we rule out the kind of behavior discussed in Young (2013), where selection alone drives wage gaps. Our model is somewhere between the work of Young (2013), in which selection is the sole driver of average wage differences, and the work of Gollin et al. (2014), where raw wage gaps are used to infer potential gains from movement.

Appendix B discusses how this basic model might be extended to account for dynamics, endogenous human capital formation, nontraded goods such as housing, and costly goods trade, and how these extensions would affect our results.

IV. Identification and Estimation

In this section, we discuss how we identify and estimate the exogenous parameters of the model $\{\Theta, \rho, q_o, w_d, \alpha_d, \tau_{do}\}$. We also note that while they are important for the counterfactuals, we do not need to take a stand on the general equilibrium parameters (γ, λ, and σ) for identification; we discuss their calibration below. We make several normalizations. First, as noted above, we assume that $\tau_{oo} = 0$ and $\tau_{do} = \tau_{od}$: movement costs are symmetric, and it is costless to live at home. Second, we normalize $\alpha_1 = 1$: because we do not observe utility levels, the only variation we have to identify α comes from people's relative preferences for locations.

[18] Note that this does not imply that average wages of people in a particular destination labor market are not affected by w_d. Average wages differ across origin, with people born in more productive locations having higher average wages.

Third, we normalize $q_1 = 1$: we identify only relative qualities of human capital generation. This normalizes productivity w_d as well: the wage w_d is what would be earned by someone living at location d who was born in location 1 and who has a skill draw of 1. This means that any aggregate improvement in human capital generation would be captured in productivities, w, and changes in q would capture changes in the spatial allocation of human capital production possibilities. Appendix B discusses identification challenges that arise in a richer model.

A. *Identification of Model Parameters*

1. Fréchet Parameters: $\{\theta, \rho\}$

Taking the log of equation (6), we have

$$\ln(\overline{\text{wage}}_{do}) = \underbrace{\ln(\bar{\Gamma}) + \ln(w_d)}_{\text{destination fixed effect}} - \frac{1}{\theta}\ln(\pi_{do}) + \underbrace{\ln(q_o)}_{\text{origin fixed effect}} + \ln(\epsilon^w_{do}). \quad (7)$$

That is, after controlling for origin and destination fixed effects, the elasticity of the average wage with respect to the proportion of migrants identifies the Fréchet parameter θ. Intuitively, if people are very similar (or destinations differ little in their skill needs), then θ is high, so the marginal migrant is not greatly less skilled than the previous migrant, and the average wage will change little with movement. However, if dispersion in talent is large (or there are large differences in the skill needs in different destinations), then the marginal migrant is much less skilled than the previous migrant, and so the wage is significantly lower.

Inspection of equation (4) shows that the error term ϵ^w_{do} also enters the definition of π_{do}. This is intuitive; any random variation that means wages for those from origin o are relatively high in destination d will encourage migration between the two locations. This correlation between the error term and the regressor π_{do} creates an endogeneity problem that will lead us to underestimate the extent of selection by overestimating θ.

We address this endogeneity concern with an instrumental-variables strategy motivated by our model. We wish to isolate the variation in π_{do} that is driven by variation in the relative amenity of d and productivity in other locations $\neg d$. The proportion of people from other origins $\neg o$ who migrate to destination d is affected by these factors, but not by the random error ϵ_{do}. The set of migration proportions $\{\pi_{d\neg o}\}$ are therefore valid instruments for π_{do}, although the first-stage relationship between $\ln\pi_{d\neg o}$ and $\ln\pi_{do}$ is nonlinear. Therefore, we follow the advice of Angrist and Pischke (2009) and instrument $\ln\pi_{do}$ with the fitted value from a "zero stage" regression in which $\ln\pi_{do}$ is regressed on a polynomial in

$\ln \pi_{d\neg o}$. Monte Carlo estimates based on a roughly calibrated version of our model, which we discuss in appendix C, confirm that this strategy leads to unbiased estimates and suggests that there are few efficiency gains to increasing the polynomial beyond a quadratic.

To separate dispersion and correlation, we use a property of the Fréchet distribution that implies

$$\frac{\text{var}(w_{do})}{(\overline{\text{wage}}_{do})^2} = \frac{\Gamma(1 - 2/\theta(1 - \rho))}{[\Gamma(1 - 1/\theta(1 - \rho))]^2} - 1. \tag{8}$$

Using data on the distribution of wages, combined with the θ identified as above, this equation identifies ρ, the parameter defining the within-person correlation of skill. Intuitively, if there is little correlation in skill types, so that everyone has some destination in which he or she excels, then the within–destination-origin pair wage variance will be low. If, in contrast, ρ is high, then people of many different skill levels will find the same location to be best, and so the variance in observed wages will be high relative to the mean.

2. Location Characteristics Affecting the Wage: $\{w_d, q_0\}$

Considering again equation (7), with the estimates of ρ and θ in hand, we can identify w_d from the destination fixed effect by noting $\bar{\Gamma} = \Gamma(1 - 1/\theta(1 - \rho))$, which is identified. We identify w_d in levels using the normalization that $q_1 = 1$. Intuitively, after controlling for selection through π_{do} and the quality of human capital through q_o, any differences in wages between locations must be driven by differences in productivity. The quality of the human capital environment q_o can be similarly determined. After controlling for productivity differences at the destination as well as selection, any differences in wages earned by people from different origins must be accounted for by the relative quality of human capital formation opportunities.

3. Characteristics Affecting Movement: $\{\tau_{do}, \alpha_d\}$

Taking the log of (4) gives a gravity equation:

$$\begin{aligned}\ln(\pi_{do}) = {} & \theta \ln(w_d) + \theta \ln(\alpha_d) + \theta \ln(1 - \tau_{do}) \\ & \underbrace{- \ln\left(\sum_j \tilde{w}_{jo}^{\theta}\right)}_{\text{origin fixed effect}} + \theta[\ln(\epsilon_{do}^{w}) + \ln(\epsilon_{do}^{\alpha})].\end{aligned} \tag{9}$$

This equation allows us to identify movement costs for each destination-origin pair that are nonparametric, in the sense that they are not functions of any other data. Intuitively, low movement could be caused by amenity differences, productivity differences, or movement costs. Among these, movement costs are the only force that leads both people from o to be unlikely to move to d and people from d to be unlikely to move to o. This intuition is confirmed by rearranging the gravity equation to give

$$[\ln(\pi_{do}) - \ln(\pi_{oo})] + [\ln(\pi_{od}) - \ln(\pi_{dd})] = 2\theta \ln(1 - \tau_{do}) + \eta_{do},$$

where η_{do} is a zero mean shock specific to the locations d and o.[19] In this equation, τ_{do} can be separately identified from η_{do} under the assumption that movement costs are symmetric, so that $\tau_{do} = \tau_{od}$. We see that movement costs are high when people tend to stay at home, and given an estimate of θ (identified as above), we can use differences in movement relative to staying at home to identify τ_{do}.

The gravity equation also allows for identification of relative amenities. In the tradition of urban economics, these amenities are residuals, not a function of any other data. The multilateral-resistance term, $\ln(\sum_j \tilde{w}_{jo}^{\theta})$, is correlated with the error, but can be removed by differencing the equation. Given this, and having identified w_d, θ, and τ_{do}, the only unknown in (9) is α_d. Intuitively, amenities are separated from movement costs because, while movement costs lead people to stay at home, amenity differences lead to a systematic flow of people toward particular locations. A location d is identified to have a high relative amenity if there is a flow of people to d that cannot be accounted for by productivity differences, measured by w_d, or by propensity to stay at home, measured by τ_{do}. We can only identify amenities up to a normalization because of the origin fixed effect in the equation.

B. Estimation

We estimate the model using Poisson pseudo–maximum likelihood (PPML). The PPML model has several advantages for estimating migration flows. First, because it estimates the level of migration, rather than the log, it can rationalize zero observed migration flows between locations. This is important because in our context, as in most studies of migration and trade flows, zero observed flows are common (Santos Silva and Tenreyro 2006). Second, the PPML model respects the general equilibrium adding-up constraints implicit in the model (Fally 2015).

[19] The term $\eta_{do} = \theta[\ln(\epsilon_{do}^{w}) + \ln(\epsilon_{do}^{\alpha}) - \ln(\epsilon_{oo}^{w}) - \ln(\epsilon_{oo}^{\alpha}) + \ln(\epsilon_{od}^{w}) + \ln(\epsilon_{od}^{\alpha}) - \ln(\epsilon_{dd}^{\alpha}) - \ln(\epsilon_{dd}^{w})]$.

Our two estimating equations, equations (7) and (9), are restated as follows:

$$\ln(\overline{\text{wage}}_{do}) = \ln(\bar{\Gamma}) + \ln(w_d) - \frac{1}{\theta}\ln(\pi_{do}) + \ln(q_o) + \ln(\epsilon^{w}_{do}), \quad (10)$$

$$\begin{aligned}\ln(\pi_{do}) = {} & \theta \ln(w_d) + \theta \ln(\alpha_d) + \theta \ln(1 - \tau_{do}) \\ & - \ln\left(\sum_j \tilde{w}^{\theta}_{jo}\right) + \theta[\ln(\epsilon^{\alpha}_{do}) + \ln(\epsilon^{w}_{do})].\end{aligned} \quad (11)$$

The identification assumption to estimate equations (10) and (11) by PPML is that the (level) error terms are mean 1 and are uncorrelated with the (exponentiated) regressors. As discussed above, we assume that the errors are mean 1, and we deal with correlation with the regressors through instrumental-variables and differencing strategies.

We proceed as follows. We first employ an instrumental-variables procedure to estimate θ. We then take this estimate of θ and estimate the system of three equations (eqq. [8], [10], and [11]) using a generalized method of moments (GMM) estimator. In implementing the procedure, we drop observations with fewer than five observed migrants from the wage data. Although our estimation method rationalizes the presence of zero observed migration between any two locations, we are concerned about small sample sizes affecting the precision of wage estimates. We bootstrap this entire procedure to generate standard errors for our estimated values of θ and ρ.

V. Estimation Results

This section presents our parameter estimates. Our main goal is to show that our structurally estimated parameters correlate with proxy measures and so they appear to measure something real. We show estimates for both Indonesia and the United States. We use our US model to estimate US-level movement costs to generate a counterfactual for a high-mobility economy. Our preferred estimates of migration cost use no structure other than symmetry. We show that this nonparametric estimate correlates with observable characteristics such as distance.

As noted above, the estimates presented in this section do not require us to take a stand on the general equilibrium parameters $\{\sigma, \gamma, \lambda\}$. These parameters will, however, be important for the results of counterfactuals presented below. We discuss the calibration and robustness of these important parameters in Section VI.

A. *Fréchet Parameters*

Table 4 presents estimates of the distributional parameters for both Indonesia and the United States. The shape parameter for the distribution of talent is determined by $\tilde{\theta} = \theta(1 - \rho)$. We estimate $\tilde{\theta}$ equal to 2.7 for the United States and 3.2 for Indonesia. Our estimate of $\tilde{\theta} = 2.7$ compares with the estimate of 2 in Hsieh et al. (2019). Separating the value into the component due to the correlation within person, ρ, and the underlying distribution, θ, we find that talent is more correlated in the United States compared with Indonesia (a value of .9 compared with .7), and shows a less disperse distribution (a value of θ of 28 compared with 13). Appendix figure 2 shows random draws from the estimated distributions for Indonesia and the United States, where each axis is the productivity level for location 1 or 2. The figure shows that, taking into account both dimensions, the skill distribution is overall more dispersed in Indonesia than in the United States.

B. *Migration Costs*

We estimate substantial migration costs. Table 4 reports the mean value of τ_{do}. On average, migrants in Indonesia must be compensated with a 39 percent higher income, while Americans require a 15 percent gain. In this sense, the United States is a high-mobility country according to our estimates.

Migration costs, for both the United States and Indonesia, correlate with distance. Figure 3 plots estimated migration cost τ_{do} against the log of distance. Particularly striking is the much lower correlation between distance and movement costs in the United States. The elasticity of cost to distance is 2 percent in the United States, compared to 15 percent in Indonesia. Several mechanisms are possible causes. It may be that transportation is cheaper in the United States. Alternatively, it may be that people

TABLE 4
ESTIMATED FRÉCHET PARAMETERS

	Indonesia (1)	United States (2)
ρ (correlation)	.74***	.90***
	(.029)	(.015)
θ (dispersion)	12.5***	27.6***
	(1.36)	(3.29)
$\tilde{\theta} = \theta(1 - \rho)$	3.18	2.69
Mean migration cost (iceberg)	.39	.15

SOURCE.—Estimates from structural estimation of model.

NOTE.—Transport costs estimated nonparametrically. Bootstrapped standard errors reported.

*** Significant at the 1 percent level.

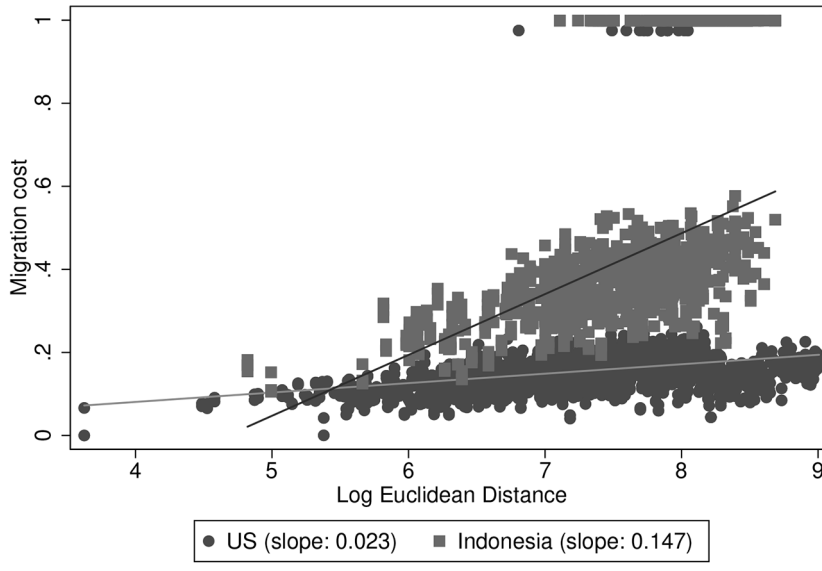


FIG. 3.—Relationship between iceberg costs and distance in Indonesia and the United States. Data from 1990 and 2010 for the United States; from 1995, 2011, and 2012 for Indonesia.

in the United States are more welcoming of migrants from physically distant communities, or that the United States is more culturally homogenous.

Measured movement costs also correlate with measures of social distance. Using census data, we construct indexes of religious and linguistic similarity.[20] Figure 4 plots the relationship between these indexes and movement costs, after controlling for distance. There is no correlation between migration costs and religion, but migration costs are statistically significantly correlated with linguistic similarity.

C. *Amenities*

Estimated amenities correlate with measured amenities. Panel A of figure 5 shows that estimated amenities are negatively correlated with the (standardized) first principal component of pollution amenities. Panel B shows that estimated amenities correlate positively with the first principal

[20] The index is constructed by calculating the probability that a person selected at random from the origin will have the same characteristic (religion or language) as a person selected at random from the destination. For example, if the origin is 50 percent Hindu and 50 percent Muslim, and the destination is 100 percent Hindu, then the religious-similarity index would be .5. If the destination was also 50 percent Hindu and 50 percent Muslim, then the index would also be .5.

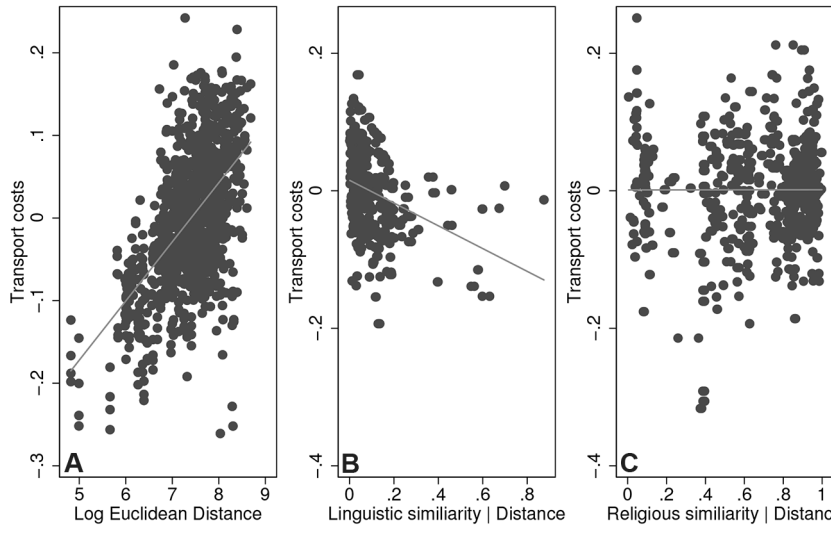


FIG. 4.—Correlates of iceberg costs in Indonesia. Data, demeaned by year, from 1995, 2011, 2012. Graphs show intensive margin transport costs (less than upper bound). *A*, Distance, $\beta = 0.072$, *t*-statistic = 31. *B*, Language, $\beta = -0.17$, *t*-statistic = −8.2. *C*, Religion, $\beta = 3.0 \times 10^{-4}$, *t*-statistic = 0.037.

component of health and market access amenities.[21] In line with the discussion of rents in appendix B.3, panel C shows that measured amenities correlate negatively with average housing costs.

D. Quality of Human Capital Formation

Figure 6 shows that the estimate of q_o (educational quality) correlates with average educational attainment. This is to be expected if people choose to receive more schooling where there are higher returns to schooling.

VI. Counterfactuals

We now turn to the policy question we posed at the start of the paper: would there be productivity gains from removing mobility constraints? We have in mind policies that change migration frictions only, and leave

[21] Appendix table 6 correlates the estimated amenities with observed amenities one by one. Each entry in the table is the regression coefficient from a separate regression of estimated amenities on each amenity. As we only have 25 estimated parameters, we do not expect individual signs to necessarily be statistically significant, but we note the general pattern in these results: overall, measures of pollution are negatively correlated with amenities; measures of health outbreaks such as malaria, tuberculosis, and vomiting are also negatively correlated with amenities, although access to health care facilities seems also to be negatively correlated; village lighting and commercial banks are positively correlated, and we see a mixed pattern for natural disasters such as flooding and earthquakes.

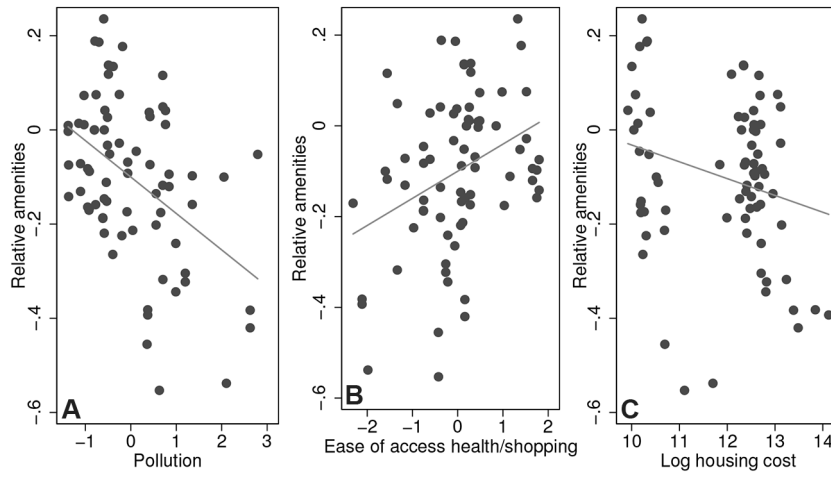


FIG. 5.—Estimated amenities against measured amenities. Correlates of estimated amenities, Indonesia. Estimated amenities from 1995, 2011, 2012. Variable is first principal component of each group of amenities. *A*, Pollution. Water, land, air, and noise pollution. $\beta = -0.077$, *t*-statistic $= -4.3$. *B*, Health/markets. Hospitals, health care, markets, and shopping centers. $\beta = 0.06$, *t*-statistic $= 3.1$. *C*, Average housing costs; sources: SUSENAS 1995, 2011, 2012. $\beta = -0.036$, *t*-statistic $= -2.1$.

all other factors, for example trade costs, unchanged. To produce counterfactuals, we need to take a stance on the general equilibrium parameters. We set these using estimates from the literature, and then evaluate the robustness of our findings to different choices.

A large literature estimates the agglomeration parameter (γ). The literature is reviewed in Rosenthal and Strange (2004) and Combes and Gobillon (2015). Recent consensus estimates suggest a γ of between 0.01 and 0.02 for the developed world, although some studies (e.g., Greenstone, Hornbeck, and Moretti 2010) suggest much higher numbers. Estimates for developing countries are more sparse and suggest a γ up to 1. We present our main estimate for $\gamma = 0.05$, but also consider robustness for numbers between 0 and 0.08. We expect that spatial integration will have a greater impact when γ is high.

A much smaller literature attempts to estimate the congestion parameter λ. As noted in appendix B, we can think of λ having a component that is due to the pure amenity spillover, λ_a, and a component that is due to endogenous changes in housing prices, λ_r. For the first term, the work in Albouy (2011) could be seen as suggesting that $\lambda_a = 0$ in the United States. In contrast, work by Combes and Gobillon (2015) suggests a λ_a of around -0.04. We take 0 as our starting point and consider various values in robustness exercises. For the housing elasticity, λ_r, there

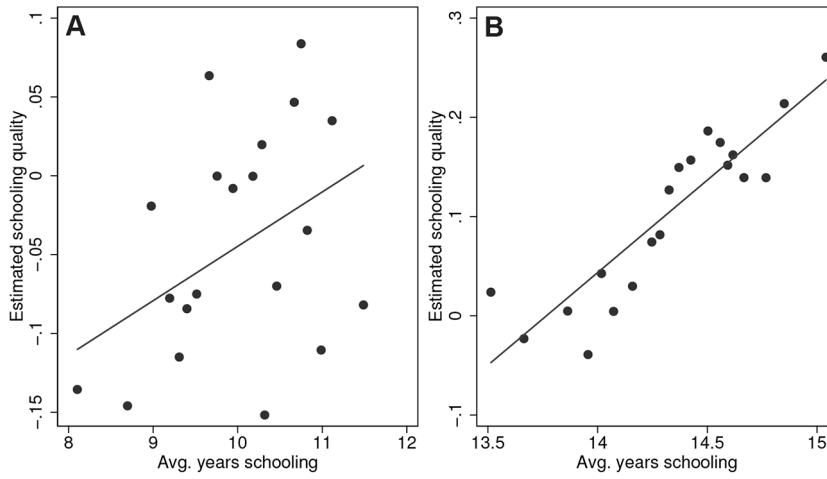


FIG. 6.—Schooling quality positively correlated with attainment (estimated schooling quality against schooling attainment). Data from 1990, 2010 for the United States; 1995, 2011, 2012 for Indonesia. Shown are binned scatterplots. Year effects are removed from both graphs. *A*, Indonesia. $\beta = 0.034$, *t*-statistic = 1.7. *B*, United States. $\beta = 0.19$, *t*-statistic = 11.

are fewer estimates available in low-income countries. In appendix table 7 we use rental data to estimate this for our sample and find a value of 0.25.[22] (For comparison, Saiz [2010] estimates this number to be 0.65 for the United States, which we take as a baseline value.) Adjusting for the expenditure share of income on housing, which we take to be 0.3, this implies $\lambda = \lambda_\alpha - 0.3\lambda_r = -0.075$. We predict that as λ decreases (and congestion becomes more important), reducing frictions will have a smaller impact. It will be hard to move people into productive areas, even if movement costs are low.

Accurate estimates of the elasticity of substitution across regions are also hard to obtain. Allen and Arkolakis (2014) use a value of 8, while Bernard et al. (2003) find a value of 4. We use 8 for our main results and consider values between 4 and 8 in robustness tables. We expect that as σ increases, there will be smaller benefits to spatial integration: a high elasticity of substitution means that the products from different locations become more substitutable, and so there are smaller costs to low production of some goods.

[22] Appendix table 7 also employs the same identification strategy to directly estimate the spillover parameters for amenities and agglomeration. We get an estimate of 0.01 for the agglomeration parameter, 0.04 for the congestion parameter, and 0.25 for the housing price parameter. While caution should be taken with these estimates, as both tests are underpowered (and the amenity test has the incorrect sign), we see these results as being broadly consistent with our choice of baseline parameters of 0.05 for the agglomeration parameter and 0 for the amenities parameter.

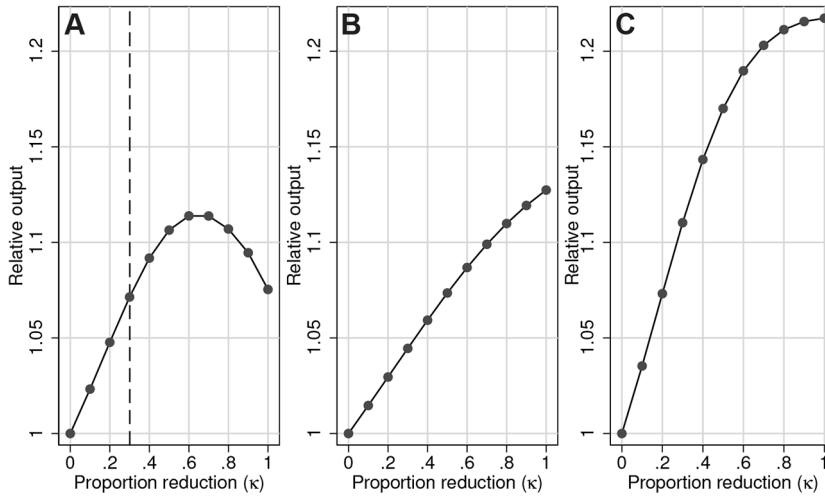


FIG. 7.—Output gains from reducing barriers to movement. Data are average across 1995, 2011, 2012 for Indonesia. The proportion reduction, κ, is defined in the text. The dashed line shows the US level of migration costs. *A*, Migration cost. *B*, Amenities. *C*, Migration cost and amenities.

A. Reducing Movement Costs

The first policy we consider is removing movement costs. On a practical basis, this might be achieved by policies such as migration subsidies (Bryan et al. 2014), migrant welcome centers, language training, and road building (Morten and Oliveira 2018). To estimate possible impacts, we scale our estimated costs by a reduction factor κ, yielding $(1 - \widetilde{\tau}) = (1 - \tau)^{1-\kappa}$, with $\kappa \in [0, 1]$. When $\kappa = 0$ this corresponds to the baseline case we estimated. When $\kappa = 1$ this corresponds to removing migration costs entirely.[23] When we undertake these counterfactuals, we allow for α_d (the combination of natural amenities and rental prices) to adjust endogenously.

We find modest gains. We predict a 7.1 percent output gain from reducing migration costs to the US level, and a 7.5 percent gain from reducing migration costs to zero. The United States is usually considered the archetype of a spatially mobile economy, so the 7.1 percent figure is probably the maximum attainable. These results are illustrated for a range of values of κ in figure 7. This figure highlights an important implication of our model: productivity effects of reducing movement costs may be nonmonotonic, and, more generally, productivity may decrease as movement costs fall. This can occur because reducing migration frictions can lead

[23] The average value of $\tau_{US} = 0.15$ and the average value of $\tau_{Ind} = 0.39$, so the policy experiment of lowering migration costs in Indonesia to the US level is equivalent to considering $1 - 0.61/0.85 = 0.28$.

workers to move away from high-productivity–low-amenity locations toward low-productivity–high-amenity locations. Our estimates suggest that, in our setting, this negative impact of reducing movement costs does not occur till costs have been substantially reduced, to lower than US levels.

These modest gains hide substantial heterogeneity across origin populations. While the average increase from eliminating all migration costs is 7.5 percent, the effect ranges from −18 percent to 68 percent.[24] That is, the people born in some provinces may see a 68 percent increase in their average wage $\Sigma_d \overline{\text{wage}}_{do}$. For a move to the US benchmark, the gains range from −5 percent to 25 percent. The distribution of gains from complete removal is depicted in panel A of figure 8, and the US benchmark is presented in figure 9. We discuss what drives these heterogeneous results in Section VI.D below.

As noted above, selection plays two roles in our model. On one hand, skill heterogeneity implies that there are gains from sorting. The greater the heterogeneity, the greater the return to sorting. On the other hand, if each additional migrant earns less than the last, selection will strongly reduce predicted gains from agglomeration. These two opposing mechanisms mean that ignoring selection could lead us to either over- or underestimate policy gains. To understand the importance of selection, we recompute productivity changes, shutting down the selection margin.[25] Sorting is the main source of output gains from removing migration costs. Column 1 in table 5 shows that all estimated gains come from improving worker sorting (we estimate a 7.5 percent gain with sorting, compared to an 8 percent loss without sorting). Ignoring selection would lead us to underestimate the gains from removing movement costs.

B. Reducing Amenity Dispersion

We consider a counterfactual in which amenities are equalized across space. This could be the result of policies such as encouraging home building in high-demand locations, which would tend to equalize rental rates (Harari 2017; Hsieh and Moretti 2019), and reducing pollution in high-productivity cities and providing equal access to schooling and hospitals, which would tend to equalize natural amenities. In undertaking these counterfactuals we assume that it is possible to fully control endogenous changes in amenity and rents so that all locations are equally desirable

[24] Recall that there is no restriction that reducing migration costs will lead to increases in output. Reducing migration costs may lead people to migrate away from high-productivity–low-amenity locations towards low-productivity–high-amenity ones. This is indeed what we see in these counterfactuals.

[25] We do this by setting the endogenous component of human capital equal to 1. This maps to a model where people are migrating based on preference shocks, such as is considered in Allen and Arkolakis (2014) and Redding (2016).

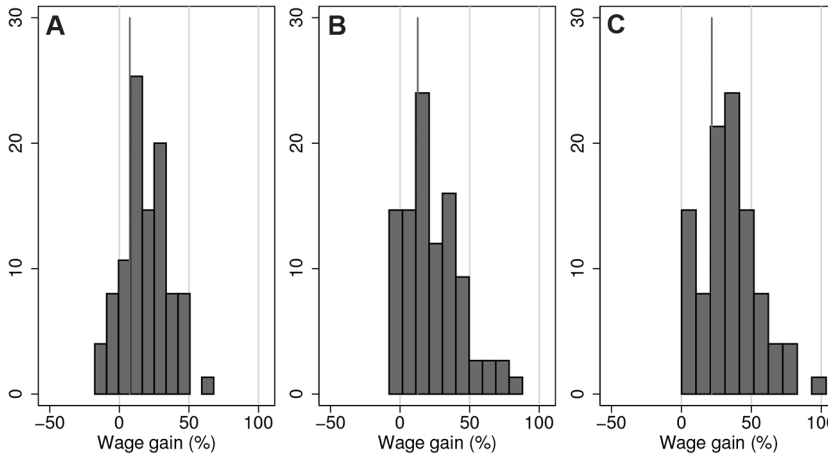


FIG. 8.—Distributional effects of fully reducing barriers to migration in Indonesia. Panels show average wage gain. The unit of observation is an origin-year. National average (weighted by population) shown by the line. A cost reduction of 100 percent is shown. Data from 1995, 2011, 2012. *A*, Migration cost. Mean, minimum, and maximum are 7.5, −17.6, 67.8. *B*, Amenities. Mean, minimum, and maximum are 12.7, −7.8, 88.0. *C*, Migration cost and amenities. Mean, minimum, and maximum are 21.7, 0.1, 103.5.

places to live. Amenities are estimated to scale. As with movement costs, we rescale amenities by a reduction factor κ, yielding $\widetilde{\alpha}_i/\alpha_1 = (\alpha_i/\alpha_1)^{1-\kappa}$, with $\kappa \in [0, 1]$. When $\kappa = 0$ this corresponds to the baseline case we estimated. When $\kappa = 1$ this corresponds to equalizing amenities across all locations.

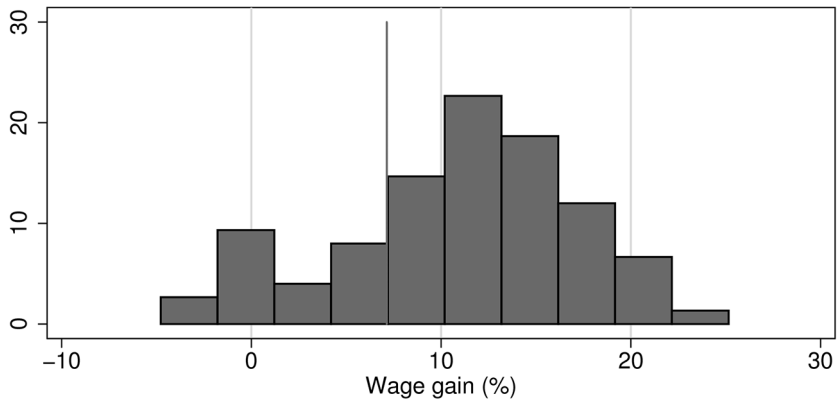


FIG. 9.—Distributional effects of reducing migration costs in Indonesia to US level. Graph shows average wage gain. The unit of observation is an origin-year. National average (weighted by population) shown by the line. A cost reduction of 30 percent is shown. Data from 1995, 2011, 2012. Mean, minimum, and maximum are 7.1, −4.8, and 25.2.

TABLE 5
OUTPUT GAIN FROM REDUCING MIGRATION BARRIERS

	Migration Cost (1)	Amenities (2)	Migration Cost and Amenities (3)
Baseline	1.075	1.127	1.217
No selection	.914	1.127	1.133

NOTE.—Table shows the output gain from removing the barrier completely. Data from 1995, 2011, 2012 for Indonesia. No selection recalculates the output gain, shutting down the role for comparative advantage.

Here we do not compute a US benchmark; this is for two reasons. First, we believe that it is plausible to have zero amenity differentials: there is no obvious reason why some locations have to have fewer services and more pollution. Second, in line with the general argument in Hsieh and Moretti (2019), we find that the United States has greater amenity dispersion than Indonesia. Hsieh and Moretti (2019) argue that that restrictive housing policies lead to high rents in some very productive locations; this would show up in our estimates as high amenity dispersion.

We find that equalizing amenities would lead to an increase in output of 12.7 percent. These gains are illustrated in panel B of figure 7. As with migration costs, we find substantial heterogeneity. Some origin locations receive wage gains of up to 88 percent. Again, we explore what drives this heterogeneity in Section VI.D below.

As above, we ask how these results are affected by selection. We find in column 2 of table 5 that in contrast to migration costs, removing the selection margin has very little effect on predicted gains. That is, by ignoring selection, we overestimate the gains from agglomeration.

C. *Reducing both Migration Costs and Amenity Differentials*

Finally, we consider eliminating both barriers—migration costs and compensating differentials—simultaneously. These gains are illustrated in panel C of figure 7. Doing so leads to a 21.7 percent output gain. The effect of reducing both barriers is slightly smaller than the sum of their independent effects, suggesting the policies are very mild substitutes. Under the policy of reducing all barriers to mobility, the origin that benefits the most would face wage increases of 104 percent. For this combined policy, accounting for selection is also important. Column 3 in table 5 shows that if we do not account for selection, we understate gains by 40 percent.

D. *Understanding Heterogeneity*

What explains why some origins gain more than others? The regions that gain the most will be those locations that have the largest ex ante frictions,

that is, the locations that are isolated due to migration costs or because they have higher amenity or low house prices. Formally, we can derive an intuitive expression that shows which locations gain the most. In the absence of any frictions (amenity differentials or movement costs), wages should be equalized within origin. That is, average wages of people from origin o who live in destination d, $\overline{\text{wage}}_{od}$, should be the same as the wages of those from origin o who live in destination d', $\overline{\text{wage}}_{od'}$. Define $\text{wage}_o = \Sigma_d \overline{\text{wage}}_{od}$ as the observed earnings of all people from origin o, and define $\widehat{\text{wage}}_o = \Sigma_d \widehat{\overline{\text{wage}}}_{od}$, the counterfactual earnings if all distortions were removed. We can show that if price adjustments are ignored,

$$\frac{\widehat{\text{wage}}_o}{\text{wage}_o} = \frac{L_o\left(\sum_d \pi_{do}\left(\overline{\text{wage}}_{do}\right)^{\theta}\right)^{1/\theta}}{L_o\left(\sum_d \pi_{do}\overline{\text{wage}}_{do}\right)}.$$

In words, the ratio of optimal wages to current wages is higher the greater the "dispersion" in averages wages across destinations. As in Hsieh and Klenow (2009), the equation makes clear what the appropriate measure of dispersion is: it is the geometric mean of wage calculated with respect to θ. This gives a simple data-driven measure of the locations that are likely to gain most.[26] We show in appendix figure 3 that the average wage gain at the origin is indeed increasing in this measure of the initial variance of wages at the origin.

E. Robustness

This section discusses four robustness exercises.

1. Agglomeration, Congestion, and Substitution Parameters

The main results use our baseline parameters for the agglomeration, congestion, and substitution. We undertake robustness over these parameters. Results are reported in appendix tables 8 through 11. As expected, when agglomeration is high, congestion forces are low, and when the

[26] This measure does not take into account general equilibrium effects through changes in p_d. If we wish to do this, we have to generalize the above equation slightly:

$$\frac{\widehat{\text{wage}}_o}{\text{wage}_o} = \frac{L_o\left(\sum_d \pi_{do}\left(p_d/\hat{p}_d\right)^{\theta}\left(\overline{\text{wage}}_{do}\right)^{\theta}\right)^{1/\theta}}{L_o\left(\sum_d \pi_{do}\overline{\text{wage}}_{do}\right)},$$

where p_d is the distorted set of prices across destinations, and $\hat{p}_d$ is the undistorted set of prices. Consider an origin location that has equal wages everywhere, except one location d^*, where wages are very high, implying large distortions. Removing all distortions, this origin location will send many workers to d^*, but this process will tend to depress prices, and hence wages, in d^*. The term $\left(p_d/\hat{p}_d\right)^{\theta}$ accounts for this effect. While it is possible to derive a closed-form solution for this number, it does not add clearly to the intuition.

elasticity of substitution is low, the gains to removing barriers to mobility increase. For the experiment of reducing both migration costs and amenities, our baseline estimate was an increase in output of 21.7 percent. The range of results in appendix table 11 is from 15.9 percent to 24.7 percent.

2. Self-Employment versus Wage Work

As noted above, a limitation of the SUSENAS and SUPAS data is that they do not record earnings for the self-employed. This may be a source of bias in our estimates. For example, if migrants are more likely to engage in wage work, our estimates of the impact of migration on average wages will include both a sample selection effect and a causal effect. This will tend to bias our estimates of the key selection parameter θ. To explore the importance of this issue, we make use of the IFLS data, which record income for both wage and self-employed work. We cannot use the IFLS for our main analysis because it is too small a sample, and we do not expect to get the same estimate of θ using the IFLS, because it is not representative of Indonesia as a whole. Nevertheless, we can use the IFLS to understand the likely direction of bias in our estimates.

Appendix table 5 replicates our motivating facts using the IFLS sample. The top panel shows results for the sample as a whole, and the bottom panel shows results for wage workers only. The table shows that the elasticity of average wage to the share migrating (col. 3), which is the inverse of the selection parameter θ, is larger for all individuals than for wage employees only. We learn two things from this exercise: (1) the motivational facts are qualitatively robust to including self-employed individuals, but (2) the dispersion of talent may be smaller for wage employees than self-employed individuals. The implied θ for all individuals is around half the size of that for wage earners alone. If this is the case, then our exercise is likely a lower bound on the gains from removing migration barriers in Indonesia as a whole. We show that this is indeed the direction of the bias in a robustness exercise in appendix table 12, where we simulate our model assuming that θ is half the size of the baseline estimate of θ. The smaller θ suggests total gains from removing migration barriers that are on the order of 23 percent, rather than the 22 percent from our baseline model.

3. Asymmetric Migration Costs

As discussed earlier, identification of the model relies on the assumption that movement costs are symmetric. One may be concerned that this is a strong assumption; for example, it would be reasonable to have a prior belief that moving from a small country town to a large metropolis may be a more costly move than the reverse. We are able to introduce asymmetry

to a limited extent by parameterizing a deviation from symmetry. For example, we can assume that $\tau_{do} = \kappa\tau_{od}$ whenever d has a higher population than o, and we can then estimate κ. We show results from such an approach in appendix table 13, where it appears that this particular parameterization leads to an increase in the aggregate gains from reducing migration costs. We do not present these larger gains as our main results and wish to urge caution in interpretation. While this particular parameterization leads to an increase in predicted gains, there may be alternative parameterizations that could lead to a decrease in predicted gains. Because we cannot accommodate all possibilities, we simply note that the size of the gains is subject to uncertainty.

4. Human Capital

As discussed in appendix B.2, endogenous human capital acquisition is a concern for our counterfactuals. We show in the appendix that it is possible to incorporate endogenous human capital into the model; doing so transforms the key sorting parameter, θ, into $\theta(1 - \eta)$, where η is the elasticity of human capital with respect to education spending. We take a pragmatic approach to addressing the concern about whether endogenous human capital acquisition would change our conclusions by computing a lower bound estimate. To do this, we calculate counterfactual average wages in each destination removing frictions and setting $\eta = 0$. Effectively this removes all existing education and any optimization response. We find in appendix table 14 that the aggregate gains of removing all frictions fall from 22 percent to 18 percent when we restrict eduction acquisition, with similar percentage drops in gains for the other counterfactual. It is important to note that because we preclude young people who have never moved from increasing education in response to changes in frictions, this is an upper bound on the importance of endogenous human capital. Overall, the results suggest that there are still substantial gains to removing migration barriers even in the presence of frictions that limit reoptimization of education.

VII. Conclusion

Large spatial wage gaps and recent experimental evidence suggest that there may be important productivity gains from encouraging internal migration in developing countries. We estimate the size of the aggregate gains in Indonesia. Our approach entails using movement data to identify constraints on migration, then using wage data to consider how removing these constraints would affect locational choices and wages, taking into account selection and general equilibrium effects.

We implement our approach using unique data from Indonesia that record location of birth, current location, and current earnings. Combined with our model, these data allow for particularly transparent identification of key model parameters. In particular, we are able to identify the key distributional parameter that determines the importance of sorting from a simple linear regression of the origin-destination wage on the origin-destination migration share.

We find aggregate output gains that are small but important, on the order of 20 percent. These estimates hide a great deal of heterogeneity, with some more constrained areas seeing gains of over 100 percent. Failure to account for selection would lead to an underestimate of the gains; accounting for selection both reduces estimated gains to agglomerating workers in one location and allows for larger gains through improved sorting. We find that the latter effect dominates.

Future research could aim to deepen our understanding of the mechanisms through which migration affects productivity. Theoretical and macroeconomic research could concentrate on the dynamic effects of encouraging migration. Microeconomic experimental evidence on the extent and nature of selection among internal migrants, as well as the strength of comparative advantage effects, would also add to our understanding. Experimental research along these lines is currently taking place as part of the broad research agenda motivated by Bryan et al. (2014) and related work, including this project.

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